

1. What is CRUD in FastAPI?

- CRUD stands for Create, Read, Update and Delete.
- It is mainly used for the basic operations to handle data in an application.
- These operations are performed using HTTP methods like POST, GET, PUT, and DELETE.

2. Why do we use the POST method for inserting data?

- The POST method is used to insert or create new data in an application.
- It sends information from client side to server side.
- POST is commonly used in forms, registration pages, and APIs to add new records like students, users, or products.

3. Difference between PUT and PATCH

- **PUT** and **PATCH** both are used to update data in an API, but they work differently.
- The **PUT** method is used to update the entire data.
- In **PUT**, you must send all the data again, even if you want to change only one thing. It replaces the old data completely with the new data.
- The **PATCH** method is used to update only specific fields. It changes only the required data without affecting the remaining fields.
- **PUT** updates the complete student details.
- **PATCH** updates only one field, such as marks or course.

4. What is the path parameter?

- A path parameter is a value passed in the URL to identify a specific item.
- It is used to get, update, or delete particular data.
- **Example:**
http://127.0.0.1:8000/test/1 here “1” is the path parameter.

5. What is the request body in FastAPI?

- In FastAPI, the request body is mainly used with POST and PUT methods to add or update data.
- The data is usually sent in JSON format and contains the information required by the application.
- The data sent to the API by the user or client for processing.

6. Why do we use Pydantic models?

- Pydantic models are used to define and validate data in FastAPI applications.
- They help ensure that the user sends the correct data types and required fields.
- Pydantic also converts JSON data into Python objects automatically.

7. What status code is returned for successful deletion?

- The status code for successful deletion is usually **200 OK**.
- It tells the user that the requested data has been removed successfully.
- In FastAPI, this status code is returned when a delete operation works properly without any errors.

8. How can we check whether a student exists before updating?

- Before updating, we check the student ID in the student list or database.
- The program compares the given ID with existing student records.
- If the ID is found, the student details are updated.
- If the ID is not found, an error message like “Student not found” is returned.

9. What happens if duplicate IDs are inserted?

- If duplicate IDs are inserted, then more than one record will have the same ID.
- This can create confusion while searching, updating, or deleting data.
- The application may change or remove the wrong record.
- To avoid this problem, every ID should be unique.

10. How does FastAPI automatically generate Swagger UI?

- FastAPI automatically creates Swagger UI from the API code.
- It shows all API endpoints in a simple web page.
- Users can open using “/docs” in the browser and test the APIs easily.